Full Stack Assignment – 30

Assignment ID: FSD-30

Title: Incident Reporting Management System

Duration: 3 Hours

1 Objective

Design and develop an **Incident Reporting Management System** that allows users to report incidents and resolvers to manage them.
The system must enforce priority handling rules and ensure **data persistence using a database**.

2 User Roles

REPORTER

- Register and log in to the system
- Report incidents
- View own reported incidents and their status

RESOLVER

- Register and log in to the system
- View all reported incidents
- Update incident status and priority

3 Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Incident Management

- REPORTER can create incidents with priority and description
- Incident data must be stored in the database
- RESOLVER can update incident details
- Incident status must be retrieved dynamically

4 Business Rule (Critical)

⚠️ **Incident priority must not be downgraded once set.**

Any attempt to lower the priority level must be rejected.

5️⃣ Technical Constraints (Mandatory)

- All incident data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for REPORTER and RESOLVER
- Incident creation form for REPORTER
- Incident listing and update view for RESOLVER

UI design is not evaluated; rule enforcement and correctness matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8 Evaluation Note

Evaluation focuses on **state immutability, role enforcement, and end-to-end execution**.